

SORTLAND, Norway

WMO: 011670

Lat: 68.700N

Lon: 15.417E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.0	-8.7	-15.7	1.0	-7.6	-14.1	1.1	-6.7	12.9	1.0	11.2	1.2	1.6	300	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	20.4	14.1	18.1	13.4	16.5	12.9	15.5	18.2	14.4	17.0	13.6	15.9	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.2	16.1	13.4	9.6	15.3	12.4	9.0	14.5	43.3	18.2	40.5	17.0	38.2	16.0	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	7.9	DB	-11.8	24.5	1.2	2.7	-12.7	26.4	-13.4	28.0	-14.1	29.5	-14.9	31.4	(4)
(5)			WB	-12.5	17.1	1.2	1.4	-13.3	18.1	-14.0	18.9	-14.7	19.7	-15.6	20.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-1.1	-1.5	-0.5	2.5	6.5	9.6	12.5	12.1	9.3	5.1	1.7	0.0		
	DBStd	5.95	3.71	3.73	3.30	3.09	3.24	2.83	2.71	2.58	2.84	3.32	3.41	3.83		
	HDD10.0	2148	344	323	324	226	117	41	5	7	45	156	249	309		
	HDD18.3	4974	603	557	583	476	366	264	183	194	271	412	499	567		
	CDD10.0	219	0	0	0	0	9	28	82	73	24	3	0	0		
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH23.3	14	0	0	0	0	0	2	9	3	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.5	3.6	3.2	3.6	4.0	3.7	3.3	3.0	3.0	3.2	3.4		
(15) Precipitation	PrecAvg	1500	144	139	136	94	68	71	82	93	144	195	159	174		
	PrecMax	1886	298	238	247	184	137	147	190	229	311	372	285	311		
	PrecMin	1223	4	75	22	29	14	20	24	40	54	52	33	0		
	PrecStd	192	70	45	61	35	32	33	44	40	67	78	75	86		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	6.7	7.6	13.3	19.7	22.2	24.1	22.7	18.1	14.2	11.1	8.3		
		MCWB	5.0	4.9	3.8	7.8	11.4	13.3	16.4	15.3	12.3	10.2	8.2	6.1		
	2%	DB	6.2	5.5	6.1	10.2	15.8	17.7	20.5	19.9	16.2	12.5	9.2	7.2		
		MCWB	4.1	3.7	3.6	6.5	10.2	12.1	14.8	14.4	11.9	9.5	6.7	5.1		
	5%	DB	5.1	4.7	5.1	8.1	12.9	15.6	18.2	17.4	14.5	10.9	7.9	6.4		
		MCWB	3.3	3.1	3.1	5.3	8.7	11.3	14.2	13.7	11.3	8.7	5.8	4.3		
	10%	DB	4.0	3.7	4.3	6.8	11.0	13.6	16.2	15.9	13.2	9.9	6.6	5.3		
		MCWB	2.2	2.2	2.4	4.4	7.6	10.6	13.5	13.3	10.7	8.0	4.9	3.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.2	5.2	8.5	12.5	14.6	17.9	17.3	13.9	11.2	8.7	6.9		
		MCDB	6.9	5.9	6.2	12.0	18.0	19.3	21.3	19.3	16.2	13.1	9.7	7.6		
	2%	WB	4.6	4.2	4.4	7.1	11.0	13.1	15.7	15.5	12.7	10.2	7.5	5.8		
		MCDB	5.7	5.1	5.5	9.5	14.8	16.6	18.7	17.9	14.6	11.6	8.6	6.9		
	5%	WB	3.6	3.4	3.5	5.8	9.3	11.9	14.7	14.5	12.0	9.2	6.1	4.7		
		MCDB	4.8	4.4	4.8	7.7	12.1	14.4	17.2	16.8	13.6	10.6	7.4	6.1		
	10%	WB	2.4	2.2	2.5	4.7	8.1	11.0	13.8	13.6	11.2	8.2	5.0	3.6		
		MCDB	3.8	3.3	4.0	6.5	10.5	13.2	16.0	15.6	12.9	9.7	6.6	5.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.1	4.4	4.7	4.8	4.0	4.2	4.4	4.5	3.8	3.7	3.8		
		MCDBR	4.3	4.1	3.8	6.4	8.4	7.7	8.3	7.8	6.3	4.5	4.7	4.3		
		MCWBR	3.5	3.5	3.0	3.7	4.1	3.5	3.8	3.8	3.5	3.4	3.7	3.8		
	5% WB	MCDBR	4.2	4.0	3.8	5.5	7.8	6.6	7.1	6.2	5.0	4.2	4.4	4.2		
		MCWBR	3.8	3.7	3.4	3.7	4.1	3.6	4.0	3.7	3.4	3.6	4.1	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.375	0.287	0.324	0.342	0.348	0.347	0.351	0.342	0.327	0.294	0.353	N/A		
	taud		2.546	2.115	2.212	2.341	2.439	2.485	2.504	2.518	2.511	2.244	2.522	N/A		
	Ebn at Noon		75	465	688	801	842	854	837	803	714	461	64	0		
	Edh at Noon		11	46	77	91	92	91	86	76	59	39	9	0		
(44) All-Sky Solar Radiation	RadAvg		0.02	0.30	1.37	3.07	4.44	4.89	4.43	3.21	1.69	0.56	0.06	0.00		
	RadStd		0.00	0.07	0.19	0.27	0.34	0.69	0.46	0.30	0.16	0.10	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.48	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page